where d_k = `distance(u,v)` is the length of the edge, and λ_3 is the usual normalization parameter: this behaves like an attractive force, so larger values for λ_3 favor smaller distances between adjacent vertices.

For edges intersections, we can just count them and multiply the numbers of intersections by a normalization factor λ_4.

Finally, to keep vertices away from edges (if you remember, this was one of the key criteria we gave, in chapter 16, to validate embeddings) we can add this term for each vertex-edge pair:

$$\frac{\lambda_5}{g_{kl}^2}$$

where g_{kl} = `distance(`e_k`,`v_l`)` and λ_5 is another normalization factor.

This term (another repulsive force) is quite expensive to compute (the edge-vertex distance is computationally heavy, as we have seen in chapter 15), and even in the original paper is not used in the default settings of the algorithm; we'll leave it out, for now, while encouraging the reader to implement it as an exercise and then experiment with it on the examples we will show.